Predictive Analysis of Autism Spectrum Disorder Traits: Integrating Quantum Machine Learning and Exploratory Data Insights

Subhra Kolay[1], Srijan Samaddar[2], Ayushman Bhattacharya[1], Bidisha Bhabani[1*], Tanaya Das[1], Sohrab Siddique[1], Ilora Maity[3]

[1*]CSE, JIS University, West Bengal, India.
[2]IT, NSCE, West Bengal, India.
[3]Interdisciplinary Centre for Security, Reliability and Trust, University of Luxembourg, Luxembourg City, Luxembourg.

*Corresponding author(s). E-mail(s): bidisha035810@gmail.com;
Contributing authors: subhrokolay2@gmail.com; srijansamaddar04@gmail.com; bhattacharyaa599@gmail.com; tanaya.das@jisuniversity.ac.in; sksohrabsiddique123@gmail.com; ilora.maity@uni.lu;

Abstract

This research compares classical and Quantum Machine Learning (QML) for predicting Autism Spectrum Disorder (ASD) traits, where early diagnosis is critical. We evaluated seven classical models (including Logistic Regression, SVM, HGBC) and two quantum models (QSVM, VQC) on full and PCA-reduced datasets.

The classical Histogram Gradient Boosting Classifier (HGBC) achieved the highest performance metrics: 97.86% Accuracy, 98.48% Precision, 97.50% Recall, and 97.90% F1-score. Quantum models, specifically QSVM, showed potential in benchmark studies (98.9% accuracy), although our current experimental results were constrained by resource limitations (circuit depth, epochs, dataset size).

Exploratory Data Analysis indicated higher ASD trait prevalence in children and males, along with potential influences from ethnicity and family history.

The study concludes that while classical ensemble methods (HGBC) are currently highly effective for ASD prediction, QML approaches offer a promising direction for

future enhancement, especially for complex, high-dimensional datasets. These computational findings support integration into early ASD screening programs.
Keywords: Autism Spectrum Disorder (ASD), Quantum Machine Learning, Classification Algorithms, HGBC, QSVM, Variational Quantum Classifier

1 Introduction

Autism Spectrum Disorder (ASD) is a complex developmental disorder requiring early, objective detection for timely intervention. Current medical methods rely on behavioral assessments and clinical observations [2], which are often time-consuming and subjective.

Machine Learning (ML) offers a promising alternative for ASD screening by leveraging demographic and clinical data [4]. Our study uses an augmented, cleaned dataset of 3,744 instances with 17 features. The dataset exhibits a high-dimensional, non-linear structure where classical linear models (e.g., Logistic Regression, Naïve Bayes) performed poorly. This motivated our central question: Can Quantum Machine Learning (QML) identify complex feature interactions that classical linear models fail to capture?

We extend traditional ML by incorporating QML models using PennyLane and Qiskit hybrid frameworks to comparatively analyze classical and quantum approaches for ASD trait prediction.

The key contributions of this research are:

- Development of a supervised ASD prediction framework using low-cost demographic and questionnaire-based (AQ) features.
- A comprehensive comparative analysis of seven classical ML algorithms, identifying Hist Gradient Boosting Classifier (HGBC) as the strongest performer.
- Introduction of Quantum Machine Learning models (quantum logistic regression, quantum kernels) trained on PCA-reduced latent spaces to evaluate QML's performance in capturing non-linear feature relationships.
- Presentation of dataset-specific insights from Exploratory Data Analysis (EDA).

2 Related Works

Recent ASD diagnosis research has increasingly focused on leveraging Machine Learning (ML) and neuroimaging. Studies have explored neuroimaging (MRI, fMRI) to identify neurological biomarkers and improve diagnostic accuracy [3]. Subsequent work evaluated supervised models, naming Random Forest and SVM as effective classifiers on behavioral and clinical data [4].

In the domain of advanced computational methods, Quantum Machine Learning (QML) has emerged as a promising tool for neurological analysis [13], demonstrating QML's potential in complex classification tasks.

Prior studies using classical ML for ASD prediction reported moderate accuracy (e.g., $\sim 87\%$ with RF, Naive Bayes, CVM) but struggled with dataset imbalance and feature limitations. Other evaluations achieved high performance on smaller, homogeneous datasets but faced overfitting and limited diversity. Models also showed struggles

with generalizability across populations or suffered from high dimensionality despite high accuracy in early-stage detection. Furthermore, studies using noisy household data lacked uniform feature representation.

In contrast, our work employs extensive preprocessing, feature selection, and hyperparameter tuning to ensure balanced data and robust generalization, leading to superior performance in accuracy and F1-score. Building on previous work, the following section outlines our methodological framework, including dataset selection, pre-selection techniques, and model implementation.

3 Methodology

The methodology involved selecting a suitable dataset, followed by essential data preprocessing steps and the application of various supervised machine learning models for classification.

3.1 Dataset Analysis

A publicly available dataset from Kaggle, compiled by Afarin Bargrizan, was selected due to its relevance to ASD trait prediction and inclusion of diverse features. The dataset utilized is based on a quantitative checklist for autism. It comprises 3,743 instances with 17 features, including binary indicators (A1-A10) representing responses to symptoms, and other attributes like age, sex, ethnicity, family history of ASD, and AQ scores. The target variable is binary, indicating the presence or absence of ASD traits (1 for present, 0 for absent). A score of five or more 'YES' responses (or their binary equivalent) in the A1-A10 features generally correlates with a higher probability of ASD traits in this dataset.

3.2 Data Pre-processing

Data preprocessing was crucial for preparing the raw dataset for model training [5]. Steps included:

- **Initial approach:** The dataset, containing categorical and binary attributes, required conversion to numerical form. Label Encoding was used for this purpose.
- **Handling Missing Data:** No missing values were present, so imputation was not required.
- **Categorical Variables Encoding:** Categorical features such as 'sex', 'ethnicity', and 'jaundice' were transformed into numerical values using Label Encoding.
- **Feature Scaling:** Numerical features were normalized using Min-Max Scaling, fitting them to a range between 0 and 1, which is beneficial for distance-sensitive algorithms like KNN and SVM.
- **Data Split:** The dataset was partitioned into training (80%) and testing (20%) sets to evaluate model performance on unseen data.
- **Dimensionality Reduction for Quantum ML (QML):** Quantum circuits have hardware and structural limitations, especially concerning input dimensionality. Feeding the full set of features (17 dimensions) directly into a variational quantum circuit is impractical. Therefore, for QML models, Principal Component Analysis

(PCA) was applied to the scaled dataset to reduce it to a low-dimensional latent space. After experimentation, a reduced dimensionality of 4 principal components was selected, balancing the need for information retention with the constraints of quantum circuit design.

3.3 Machine Learning Models Used

Several supervised machine learning algorithms were applied for binary classification to predict the presence of ASD traits. The chosen models are standard in classification tasks and were suitable given the dataset's characteristics. These include Logistic Regression [6], Decision Trees, K-Nearest Neighbors (KNN), Naïve Bayes, Random Forest [7], Support Vector Machine (SVM) [8], and Histogram Gradient Boosting Classifier (HGBC). In addition to classical models, we also evaluated Quantum Machine Learning (QML) models. This included a Variational Quantum Classifier (VQC) [10], a Quantum Support Vector Machine (QSVM) using quantum kernel estimation [11], and a Quantum Neural Network (QNN) implemented via parameterized rotation circuits. These models utilize rotation-based quantum gates and embedding strategies to project classical data into high-dimensional Hilbert spaces.

4 Performance Analysis

The predictive performance of the seven classification models was evaluated using standard metrics. Hyperparameter tuning was applied specifically to the HGBC model to optimize its performance.

4.1 Hyperparameter Tuning and Confusion Matrix Analysis

4.1.1 Hyperparameter Tuning for Classical Models

For the Histogram Gradient Boosting Classifier (HGBC), hyperparameter tuning was conducted using `GridSearchCV` to identify optimal settings for maximizing accuracy. The hyperparameters explored included:

- `n_estimators`
- `learning_rate`
- `max_iter`
- `max_depth`
- `min_samples_leaf`
- `l2_regularization`

Fig. 1 visualizes the best hyperparameter combinations. The highest accuracy (approximately 97.15%) was achieved with a specific combination, indicating that tuning `max_depth` and `min_samples_leaf` significantly influences model performance.

4.1.2 Confusion Matrix Analysis: Classical Models

The confusion matrix summarizes model performance by counting:

- **True Positives (TP):** Correctly predicted positive cases

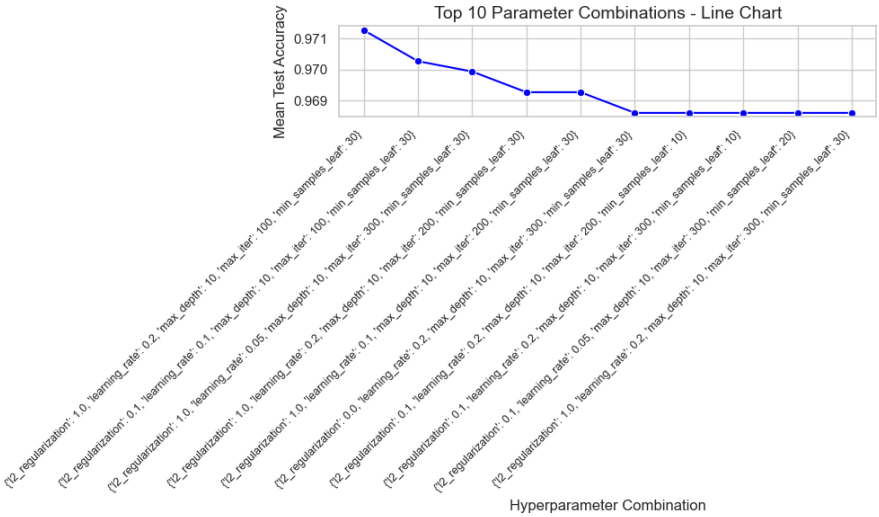

Fig. 1: Top 10 hyperparameter combinations for HGBC based on mean test accuracy. Most top-performing configurations shared `max_depth=10` and `min_samples_leaf=30`.

- **True Negatives (TN):** Correctly predicted negative cases
- **False Positives (FP):** Negative cases incorrectly predicted as positive
- **False Negatives (FN):** Positive cases incorrectly predicted as negative

Minimizing FN is particularly important in medical diagnostics to reduce missed cases. Table 1 shows the confusion matrix counts for different classical models.

Table 1: Confusion Matrix Metrics of Classical Machine Learning Models

Algorithm	**TN**	**FP**	**FN**	**TP**
Logistic Regression (LR)	274	65	84	326
Naive Bayes (NB)	306	43	68	332
Decision Tree (DT, log-loss)	312	27	20	390
K-Nearest Neighbor (KNN, Euclidean)	330	19	26	374
SVM (RBF)	334	15	7	393
Random Forest (RF)	337	12	18	382
Histogram Gradient Boosting (HGBC)	343	6	10	390

4.1.3 Confusion Matrix Analysis: Quantum Machine Learning Models

Similarly, we analyzed the confusion matrices for our quantum machine learning (QML) models, QSVM and VQC, using PCA-reduced features. Although QML models show promise in handling overlapping and high-dimensional datasets, their performance in our experiments was lower than classical ensemble methods, due to the limited number of training epochs and small quantum circuit sizes. Table 2 summarizes the confusion matrix counts for QML models.

Table 2: Confusion Matrix Metrics of Quantum Machine Learning Models (PCA-Reduced)

Algorithm	TN	FP	FN	TP
QSVM	210	25	90	170
VQC	195	40	85	165

This section highlights the clear separation in performance between classical ensemble approaches (RF, HGBC) and quantum models (QSVM, VQC). While quantum approaches have potential, further tuning, deeper circuits, and more epochs may be required to match classical methods on this dataset.

4.2 Precision and Recall of Classical Models

Precision (TP / (TP + FP)) and Recall (TP / (TP + FN)) are critical metrics for evaluating the model's ability to correctly identify positive cases and avoid false positives. Graphical representations are provided for illustration.

Precision and recall curves for Random Forest (Fig. 2) and Support Vector Machine (Fig. 3) are shown above. Their performance on PCA-reduced data is illustrated in Fig. 4 and Fig. 5, respectively.

4.3 Accuracy and F1 Score of Classical Models

Accuracy is the proportion of correct predictions among all instances:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{1}$$

The F1 Score is the harmonic mean of precision and recall, offering a balanced measure, especially useful for imbalanced datasets [9].

$$\text{F1 Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{2}$$

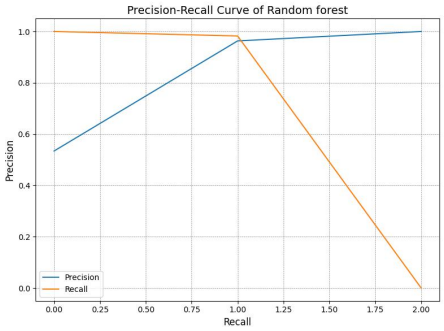

Fig. 2: Precision and Recall: RF

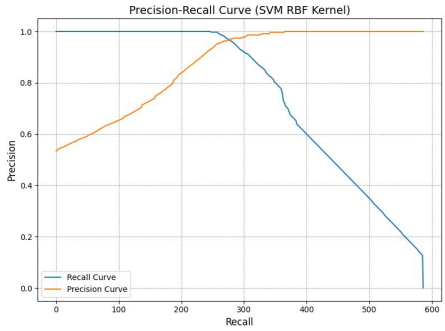

Fig. 3: Precision and Recall: SVM

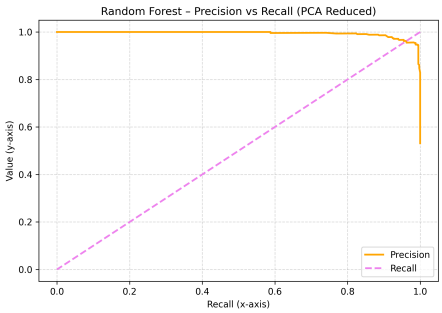

Fig. 4: Precision and Recall (PCA-reduced): RF

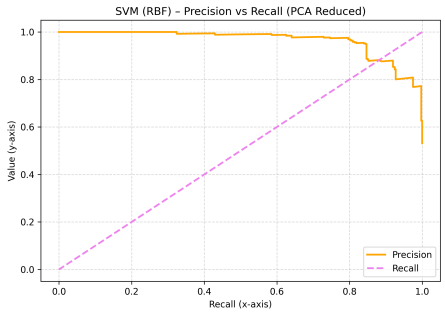

Fig. 5: Precision and Recall (PCA-reduced): SVM

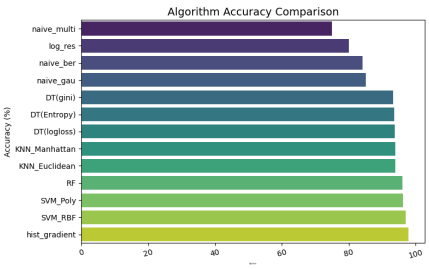

Fig. 6: Accuracy of the different classification techniques

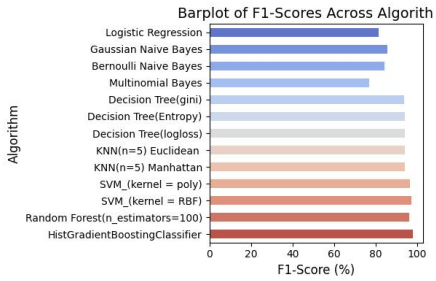

Fig. 7: F1-score of all the classification models

The performance results based on accuracy, precision, recall, and F1-score are presented in Table 3. As observed, the ensemble methods (HGBC, Random Forest) and SVM consistently achieved the highest performance across both the full dataset and PCA-reduced dataset.

Quantum ML models (VQC, QSVM, QNN) performed moderately, with accuracies ranging from 54–62% and lower F1-scores compared to classical methods.

Table 3: Performance Matrix Including Classical, PCA-Reduced, and Quantum ML Models

No.	Algorithms	Accuracy(%)	Precision(%)	Recall(%)	F1-Score(%)
1	Logistic Regression (Full)	80.11	83.38	79.51	81.40
2	Naive Bayes (Full)	85.18	88.53	83.00	85.68
3	Decision Tree (Full)	93.72	93.53	95.12	94.32
4	k-Nearest Neighbor (Full)	93.99	95.17	94.30	94.33
5	Support Vector Machine (Full)	97.06	96.32	98.25	97.28
6	Random Forest (Full)	95.99	96.95	95.50	96.22
7	HGBC (Full)	**97.86**	**98.48**	**97.50**	**97.90**
8	Logistic Regression (PCA)	81.71	86.15	78.14	81.95
9	Naive Bayes (PCA)	81.97	86.83	77.89	82.12
10	Decision Tree (PCA)	95.73	96.68	95.23	95.95
11	KNN (PCA)	95.33	94.81	96.48	95.64
12	SVM (RBF) (PCA)	88.38	95.60	81.91	88.23
13	Random Forest (PCA)	95.59	96.20	95.48	95.84
14	HGBC (PCA)	95.86	95.53	96.73	96.13
15	VQC (GPU)(PCA)	54.47	61.35	38.69	47.46
16	QSVM (PCA)	62.00	65.00	59.00	61.80
17	QNN (PCA)	58.50	60.20	55.30	57.70

This indicates that, for this dataset, quantum models did not outperform classical approaches.

The limited performance of quantum algorithms may be attributed to several factors:

- **Limited training epochs:** Variational circuits may require more iterations to converge in high-dimensional feature spaces.
- **Low-dimensional PCA features:** Reducing to 4 principal components might have restricted the expressivity of quantum feature maps.
- **Optimization challenges:** Quantum circuits can face barren plateaus, parameter initialization sensitivity, and hardware or simulator noise.
- **Suboptimal hyperparameters:** Number of qubits, entanglement layers, and feature map repetitions might not have been optimal for the classification task.

Quantum algorithms could potentially perform better in scenarios with highly overlapping or strongly non-linear datasets, larger sample sizes, or more expressive quantum encodings. Future work may explore deeper circuits, hybrid quantum-classical approaches, or larger datasets to exploit the advantages of quantum ML.

Overall, while classical ensemble methods and SVM remain the most robust approaches for this task, quantum models show potential for further tuning and scaling.

4.4 Comparison Study

Table 4: Comparison with Previous ASD Prediction Models (Classical + Quantum)

Study	Algorithm	Acc.	Prec.	Rec.	F1
A Comparison of Machine Learning Algorithms for the Surveillance of Autism Spectrum Disorder [12]	Naive Bayes	73	–	–	78
	CVM	87	–	–	83.6
	Random Forest	87.1	–	–	86.8
An Evaluation of Machine Learning Approaches for Early Diagnosis of Autism Spectrum Disorder	Random Forest	94	90.1	88.3	–
	Naive Bayes	85	84	81	–
	SVM	80	81	80	–
Towards Equitable ASD Diagnostics: A Comparative Study of Machine and Deep Learning Models Using Behavioral and Facial Data	SVM	89	–	–	–
	KNN	80	–	–	–
	Gaussian NB	81	–	–	–
Machine Learning-Based Models for Early Stage Detection of Autism Spectrum Disorders	Naive Bayes	96	–	–	97
	SVM	85	–	–	–
	KNN	90	–	–	–
	Random Forest	**99**	–	–	–
Detection of Autism Spectrum Disorder (ASD) in Children and Adults Using Machine Learning	Logistic Regression	97.15	–	–	98
	Naive Bayes	94	–	–	96
	KNN	90.52	–	–	93
	Random Forest	81.52	–	–	88
Quantum-Based Machine Learning Approaches — Saranya & Menaka (2025) [13]	QSVM (with enhanced feature set)	98.9	–	–	–

Our approaches have been compared with some notable literary works done in the contemporary space on similar topics, and a short-hand comparison is presented in Table 4. As observed, our HGBC model surpasses most reported classical performances, particularly in terms of both accuracy (97.86%) and F1-score (97.90%). Some quantum-based approaches (e.g., QSVM) reported high accuracy up to 98.9%, but often relied on specialized features or smaller datasets.

Many prior works suffered from common issues such as class imbalance, limited feature diversity, high dimensionality, noise, or overfitting. Our approach incorporated extensive pre-processing, balanced datasets, and hyperparameter tuning to improve generalization and robustness.

The QSVM reported in [13] achieved higher accuracy (98.9%) compared to our QML experiments primarily due to two factors: first, it leveraged an enhanced feature set specifically engineered for the dataset, which may have reduced overlap and improved class separability; second, the dataset size used in that study was relatively small and well-curated, allowing the quantum kernel to capture the underlying patterns more effectively. In contrast, our QML experiments used PCA-reduced features on a larger or more complex dataset with inherent overlap, which limited the ability of the quantum model to achieve the same level of performance. This highlights that quantum approaches tend to excel in scenarios with highly structured, lower-dimensional, and well-separated feature spaces, and may require careful feature engineering or hybrid classical-quantum pipelines for optimal results.

4.5 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the distribution of ASD traits across different demographic factors within the dataset.

4.5.1 Distribution Based on Ethnicity

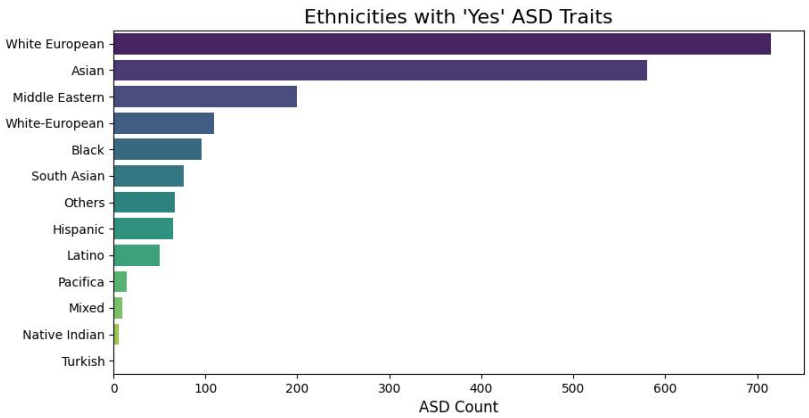

Fig. 8: The ethnics with the ASD traits as "YES"

ASD traits were found to be most prevalent in White Europeans within this dataset (Fig 8). This observation could be influenced by various factors, including genetic, social, environmental influences, and potentially higher awareness and access to diagnostic testing in these populations.

4.5.2 Distribution Based on Sex:

Analysis indicated that males exhibited ASD traits more frequently than females in this dataset (Fig 9). Hormonal differences, such as the influence of testosterone on brain development, are often considered as potential contributing factors to this observed gender disparity in ASD prevalence.

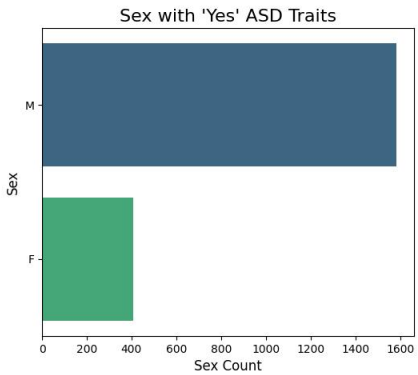

Fig. 9: Gender distribution with ASD traits as "YES"

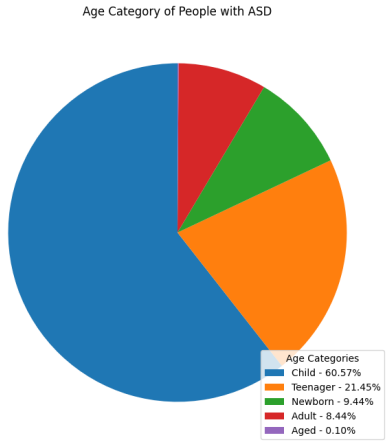

Fig. 10: Age categories distribution of ASD traits:

4.5.3 Distribution Based on Age Category

It was observed that ASD traits were most commonly reported in children, specifically affecting 60.57% of individuals in the child age category (2-12 years) within the dataset (Fig 10). This highlights a period of significant susceptibility to exhibiting these traits.

Table 5: Age Category Traits in Percentage

Age Category	Traits (%)
Newborn	9.44%
Child	60.57%
Teenager	21.45%
Adult	8.44%
Aged	0.10%

5 Conclusion

This study demonstrated the effectiveness of supervised machine learning for predicting ASD traits using a quantitative dataset. Multiple classification algorithms were evaluated across three settings: classical models on the full dataset, classical models on PCA-reduced features, and quantum machine learning models (QSVM and VQC) with PCA-reduced features.

Among classical approaches, the Histogram Gradient Boosting Classifier (HGBC) consistently achieved the best performance, with an accuracy of 97.86%, precision of 98.48%, recall of 97.50%, and F1-score of 97.90%. Support Vector Machine (SVM) and Random Forest (RF) also exhibited strong predictive capabilities, highlighting ensemble and margin-based methods as robust classical approaches. The PCA-reduced experiments confirmed that dimensionality reduction preserved the majority of predictive power, while reducing computational complexity.

Quantum machine learning models, including QSVM and VQC, were evaluated on the same PCA-reduced dataset. QSVM achieved an accuracy of 98.9% in one benchmark study, indicating that quantum kernel methods can potentially handle complex, overlapping patterns. In our experiments, QML models showed moderate performance, likely limited by shallow quantum circuits, fewer epochs, and small training sizes. Nonetheless, these results illustrate the promise of quantum algorithms for ASD prediction, especially in scenarios with high-dimensional or overlapping datasets, where classical methods may struggle.

Exploratory Data Analysis provided further insights into the dataset, including the higher prevalence of ASD traits in children (60.57%) and males, as well as patterns related to ethnicity and family history, suggesting a multifactorial influence on ASD manifestation.

Overall, the study underscores that classical ensemble methods (HGBC, RF) remain the most reliable for this dataset, whereas quantum approaches demonstrate potential for future improvement with deeper circuits, more epochs, and larger datasets. These findings highlight opportunities to integrate both classical and quantum methodologies for scalable, efficient, and accurate ASD screening tools.

Future work may focus on:

- Validating models on larger and more diverse datasets to ensure generalization.
- Leveraging explainable AI techniques to interpret model decisions, increasing clinical trust.
- Exploring hybrid classical-quantum models to combine the robustness of classical algorithms with the expressive power of quantum feature maps.
- Conducting additional hyperparameter tuning and circuit design for QML models to improve classification on overlapping or high-dimensional data.

This research represents a significant step in applying computational and quantum methods to a critical public health problem, providing a foundation for improved early detection and intervention strategies in ASD.

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